REWT

?

Learning an energy landscape for state refinement

Can revising an internal state improve an answer?

Kenju Tomita · Rochester Institute of Technology

The goal: compose facts into an answer

A small rule task makes each reasoning step checkable.

```
B>C;  A>B;  C>D;  X>Y;  from A go 3=?
```

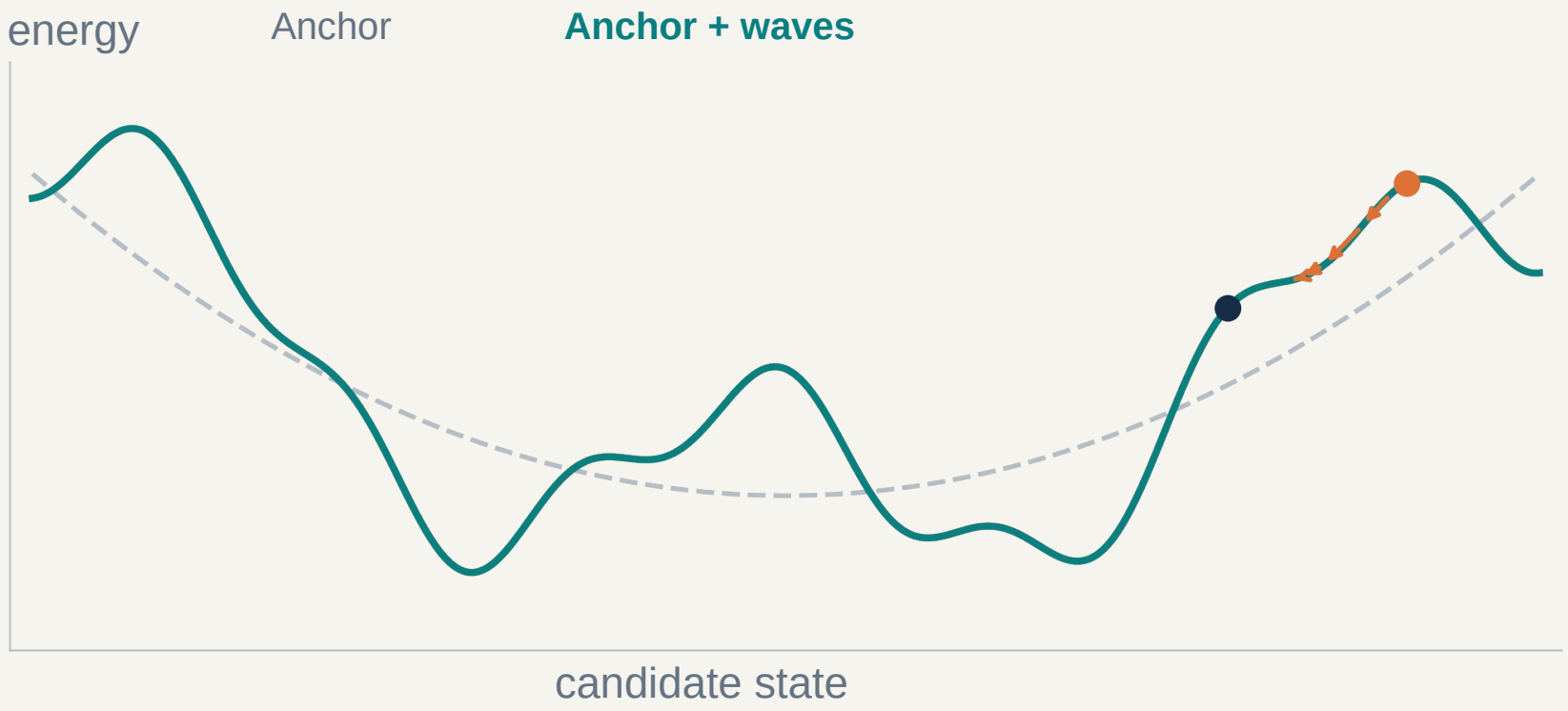


Correct trace: B, C, D **Final answer: D**

Illustrative task, not a benchmark sample. Actual tests use shuffled facts and disjoint semantic identities.

Energy is a learned compatibility score

Lower energy means the model prefers that state. It does not certify truth.



THE LOOP

Guess → revise

Follow the slope.

Accept a lower score.

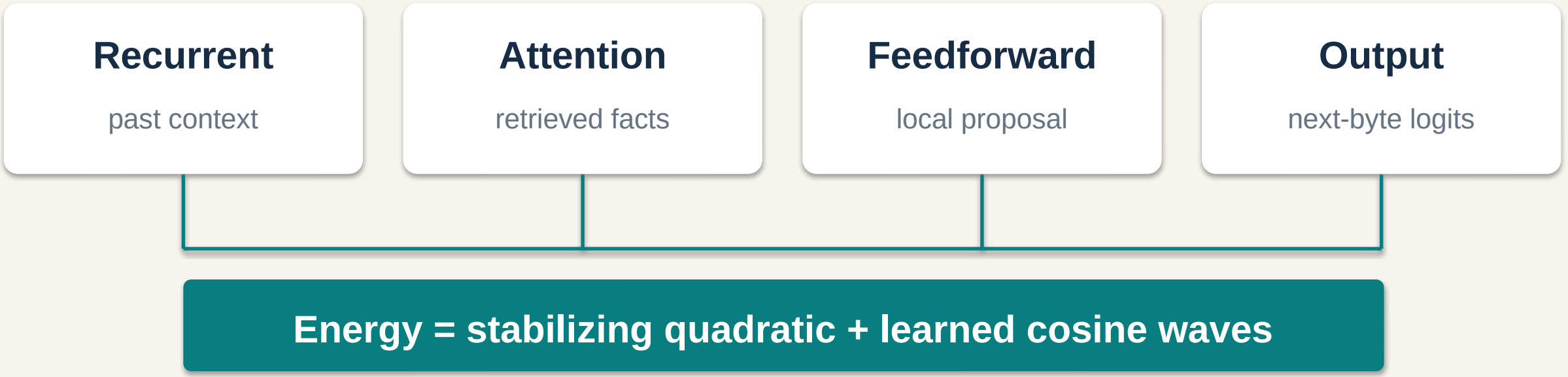
Repeat with fixed weights.

A lower local valley can still correspond to a wrong answer.

Illustrative one-dimensional landscape; the implemented state is high-dimensional. Background: Hoover et al., 2023.

One energy couples several state blocks

Observed text proposes an initial state; refinement updates the joint workspace.



Quadratic: stay near a context-informed proposal. Waves: learn coupled preferences.

Implemented language-model core. “Wave” means Fourier components in state space, not sound or a quantum mechanism.

Training shapes the landscape. Inference moves the state.

TRAINING

Change parameters θ

Use labeled examples.
Differentiate through refinement.
Update weights with AdamW.

INFERENCE

Change candidate state s

Read the supplied context.
Take accepted state updates.
Keep learned weights fixed.

$$s \leftarrow s - \eta \nabla E\theta(\text{context}, s)$$

State optimization and parameter optimization are separate. Fixed-weight and analytic-gradient checks are automated.

Three questions need three different measurements

01 · NUMERICS

Does energy fall?

Check accepted updates and analytic gradients.

02 · QUALITY

Is the answer right?

Score the complete trace and final answer.

03 · COST

What does it cost?

Measure training and inference;

keep compute budgets visible.

Disjoint problem identities · Validation selection · Three seeds · Frozen test models

Protocols and complete outputs are saved in reports/rewt_architecture_20261022 and reports/rewt_reasoning_20261026.

Learned overwrite memory survives longer gaps

Return the latest value after a key has been written twice.

K=2 … K=9 … query K → 9



At a 191-byte gap: 9.4% local → 83.9% delta

Oct. 22 matched ablation: 3 seeds, 384 prompts, 1,536 updates; 32,752 vs 34,321 parameters. Narrow key/value grammar.

Lower energy did not reliably fix wrong answers

Removing refinement from frozen language-model checkpoints did not reduce complete-answer accuracy.

Complete-answer accuracy	Normal rounds	Zero rounds
GRU control	**34.16%**	**34.98%**
Delta variant	**35.39%**	**35.39%**

The old loop optimizes the same landscape without rereading the facts.

Oct. 22 validation-only intervention: 3 seeds per variant, 162 prompts/model; includes arithmetic, rules and recall.

Each predicted state drives another memory read

A learned byte encoder supplies keys, values, the starting query and the read count.

READ 1

query A → retrieve B

refine this read's state

READ 2

query B → retrieve C

refine this read's state

READ 3

query C → retrieve D

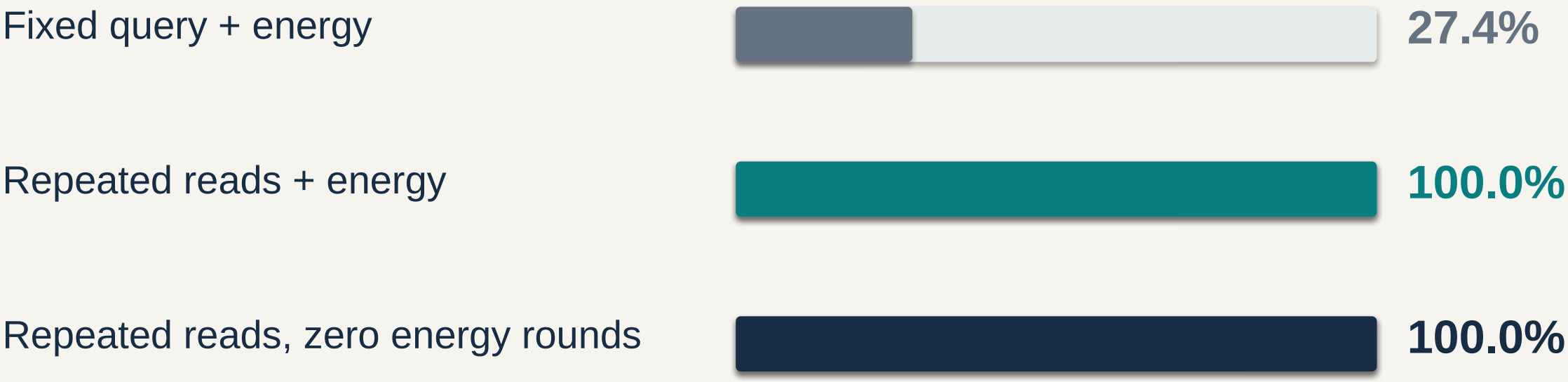
refine this read's state

Input: raw bytes **Output: predicted nodes + predicted length**

No symbolic parser or solver at inference. Each read defines a new energy; descent is guaranteed only within that read.

Repeated reads help; the wave benefit is still unproven

Complete trace + answer on 960 fresh rule prompts, across three cold starts.



Both repeated-read arms also solve 480/480 prompts with eight distractor edges.

68,502 allocated parameters; 6,000 training identities; 800 updates/fit. Same batches and labels; zero-round arm has 60,950 active parameters.

A useful result with a clear boundary

DEMONSTRATED

Bounded rule composition

One to three hops.
Known letter vocabulary.
Shuffled facts and more distractors.

STILL OPEN

Broader reasoning

Natural language and arithmetic.
Longer or branching programs.
A causal benefit from wave refinement.

Wording is still fragile: 43.1% on the two withheld validation phrasings.

The 100% test score is limited to its specified templates. Validation contains two different withheld templates and remains weaker.

TAKEAWAY

Make refinement earn its computational cost

1. **Teach correction** — Reward actual answer repairs
2. **Integrate rereading** — Bring the loop into the language model
3. **Compare fairly** — Measure quality at matched compute

Memory and repeated reads help. Lower energy alone is insufficient.

Research next steps; these are not completed general-language results. Questions welcome.

The within-read descent contract

For one read, the context-dependent coefficients stay fixed.

$$E(\delta) = \tfrac{1}{2}\,\delta^{T} P\,\delta + \Sigma\, a_k \cos(\omega_k^{T}\delta + \varphi_k)$$

δ = candidate − anchor P is positive diagonal

$$\delta' = \delta - \eta\,\nabla E(\delta) \qquad \text{accept only if } E(\delta') < E(\delta)$$

A new memory read changes the anchor and amplitudes. Do not compare energies across reads.

Formula for the new structured head. The earlier language-model core also has quadratic compatibility terms.

The plain read loop is sufficient on this test

All arms: three seeds, identical 800-update batches and supervision.

Arm	Active parameters	Training CPU / fit
Fixed query + energy	**68,502**	**38.3 s**
Repeated reads + energy	**68,502**	**31.5 s**
Repeated reads, zero energy rounds	**60,950**	**22.5 s**

Frozen intervention: disable energy → 100%; freeze queries → 33.3%.

Single-thread process time on a shared CPU host, not a GPU/FLOP comparison. Setup, validation and I/O excluded from fit timing.

Ideas are credited; project results are local

Vaswani et al. · Attention Is All You Need · 2017

Hoover et al. · Energy Transformer · 2023

Yang et al. · Parallelizing Linear Transformers with the Delta Rule · 2024

Project evidence

Architecture, memory and numerical audits · October 22

Learned rereading, controls and raw outputs · October 26

Local source reports: reports/rewt_architecture_20261022/FINDINGS.md and reports/rewt_reasoning_20261026/FINDINGS.md